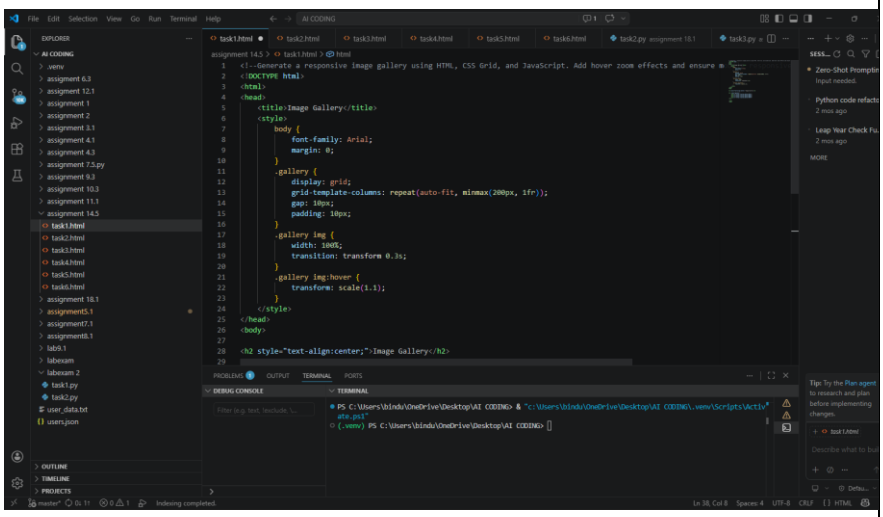
| SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE | | DEPARTMENT OF COMPUTER SCIENCE ENGINEERING | |
|---|---|---|---|
| **Program Name:** B. Tech | | **Assignment Type: Lab** | **Academic Year:**2025-2026 |
| **Course Coordinator Name** | | Dr. Rishabh Mittal | |
| **CourseCode** | 23CS002PC304 | **Course Title** | AI Assisted Coding |
| **Year/Sem** | III/II | **Regulation** | R23 |
| **NAME:** | A.Rithika | **HALLTICKENT NUMBER** | 2303A54048 |
| **Duration** | 2 Hours | **Batche** | 48 |
| **Assignment Number: 14.5** | | | |

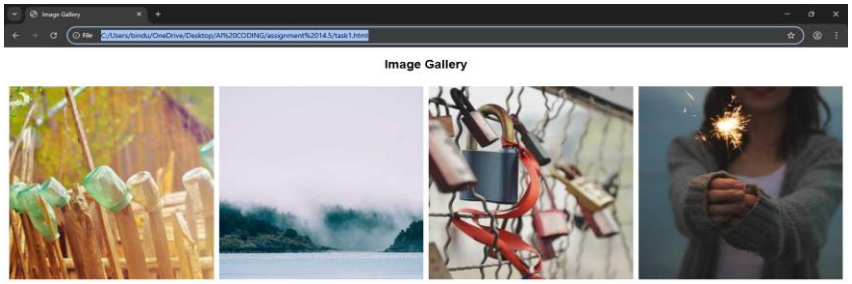
| Q.No. | Question | *Expected Time to complete* |
|---|---|---|
| 1 | **Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation**<br>**Lab Objectives**<br>• Design **functional, visually appealing** web applications using HTML, CSS, and JavaScript with AI assistance.<br>• Apply **responsive, accessible, and interactive design principles**.<br>• Create **practical UI components** for real-world web applications.<br>• Use AI to optimize **layout, UX, and performance**.<br>---<br>**Task 1 – Image Gallery Website**<br>Create an image gallery web page.<br>**Scenario**<br>Create a responsive image gallery using HTML, CSS Grid, and basic styling. The gallery should adjust to different screen sizes and include hover effects.<br>**Prompt Used**<br>Generate a responsive image gallery using HTML and CSS Grid | Week7 - Friday |

with hover zoom effects and mobile responsiveness.

**CODE:**




**OUTPUT :**

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**Explanation**

This program creates an image gallery using CSS Grid. The images automatically adjust based on screen size. A hover effect is added to make the images zoom when the user points at them.

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**Task 2 – Profile Card Generator**

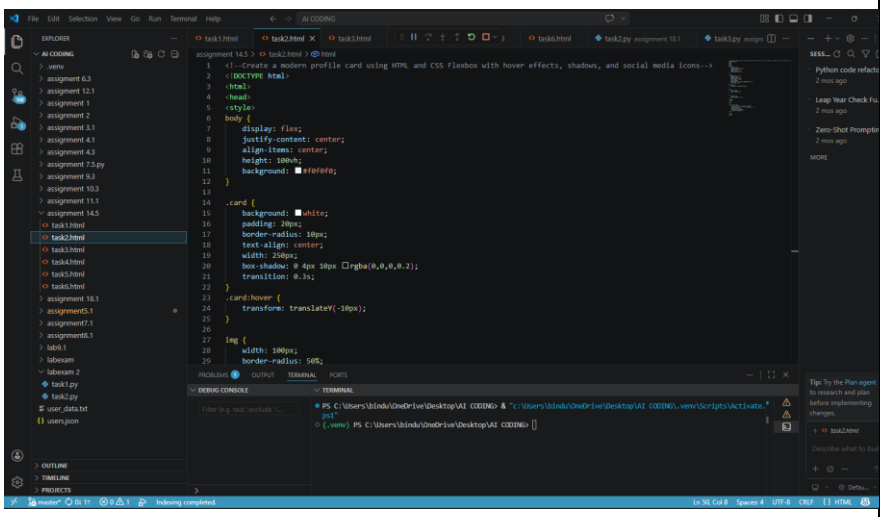
**Scenario**

Design a profile card using HTML and CSS with hover effects and user details.

**Prompt Used**

Create a modern profile card using HTML and CSS Flexbox with hover effects and shadow styling.

**Code**

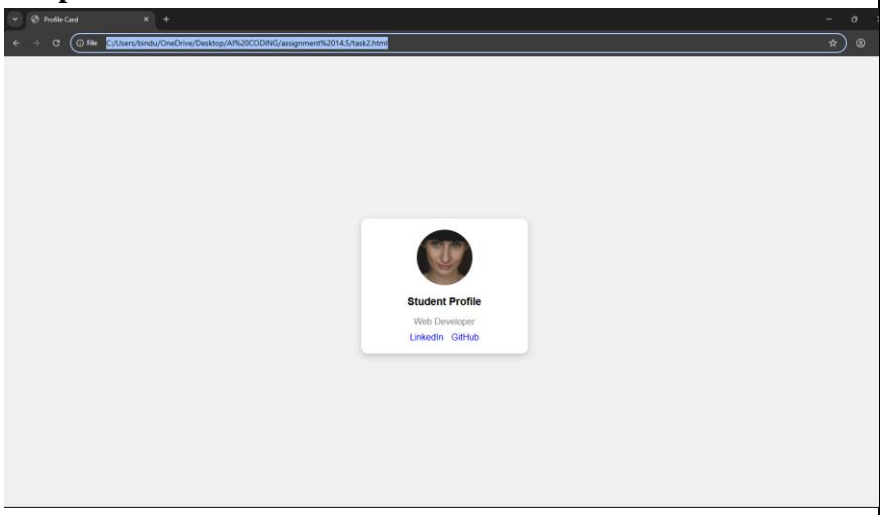


```html
<!--Create a modern profile card using HTML and CSS Flexbox with hover effects, shadows, and social media icons-->
<!DOCTYPE html>
<html>
<head>
<style>
body {
    display: flex;
    justify-content: center;
    align-items: center;
    height: 100vh;
    background: #f0f0f0;
}

.card {
    background: white;
    padding: 20px;
    border-radius: 10px;
    text-align: center;
    width: 250px;
    box-shadow: 0 4px 10px rgba(0,0,0,0.2);
    transition: 0.3s;
}
.card:hover {
    transform: translateY(-10px);
}

img {
    width: 100px;
    border-radius: 50%;
```

**Output:**




**Explanation:**

This program creates a simple profile card. CSS is used to center the card and add shadow effects to make it look attractive.

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**Task 3 -Develop a dynamic shopping cart system.**

**Requirements:**

**Scenario:**

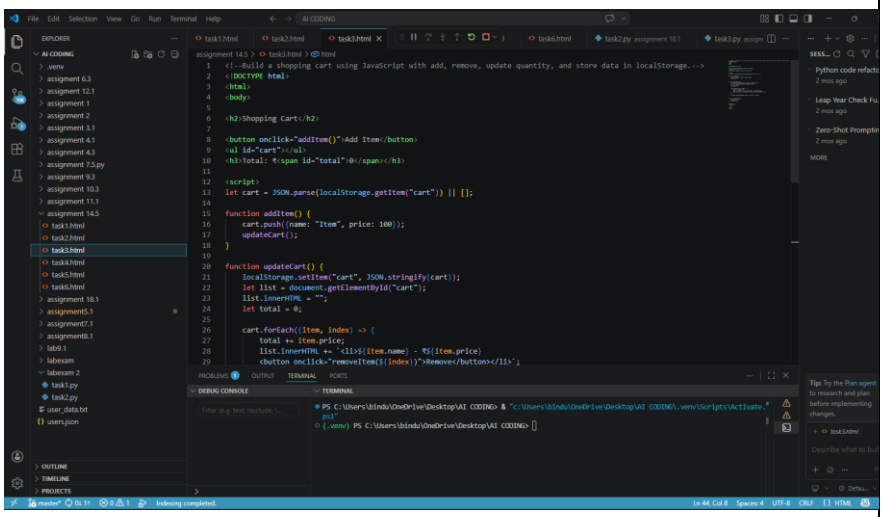
Create a simple shopping cart using JavaScript with add and remove functionality.

**Prompt Used:**

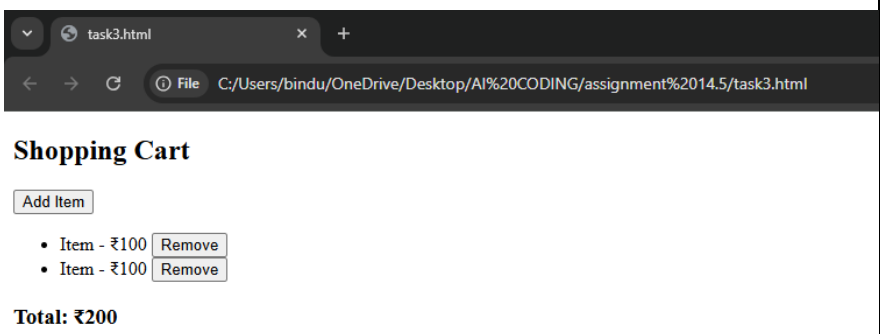
Build a shopping cart using JavaScript with add and remove item

functionality and display total price.

**CODE:**




OUTPUT:




**Explanation**

This program uses JavaScript to store items in an array and display them dynamically on the web page.

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**Task 4 – Role-Based Access Control Application**

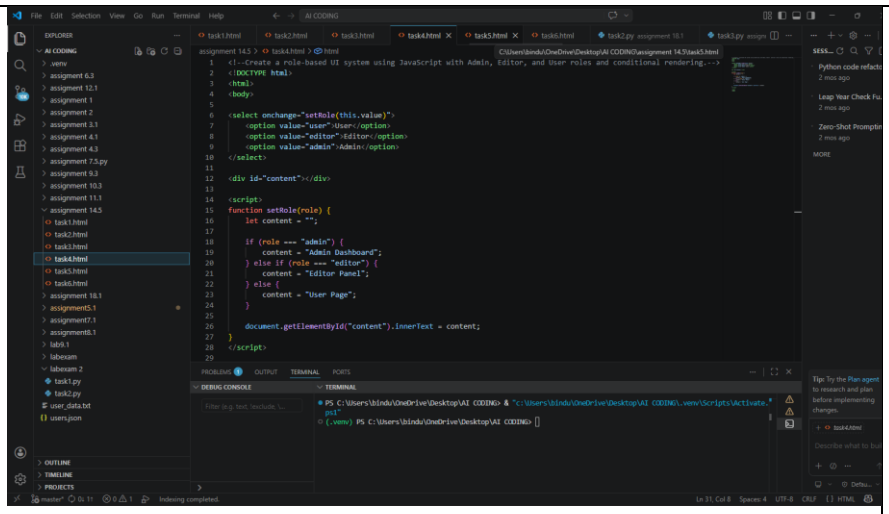
**Scenario**

Create a system where content changes based on user role (Admin/User).

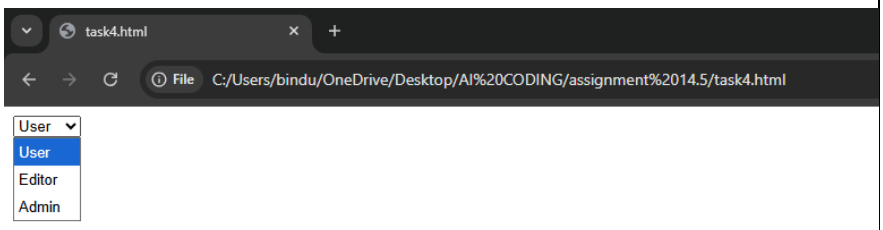
**Prompt Used**

Create role-based UI using JavaScript with conditional rendering for admin and user roles.

**CODE:**




**OUTPUT:**




**Explanation**

The program checks the selected role and displays content accordingly using conditional statements.

## Task 5 – Multi-Language Website

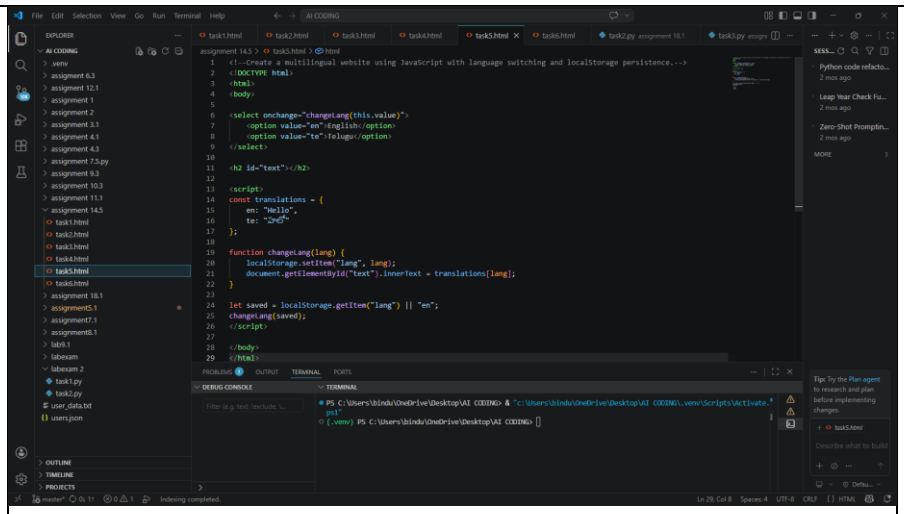
**Scenario**

Create a website that supports multiple languages.

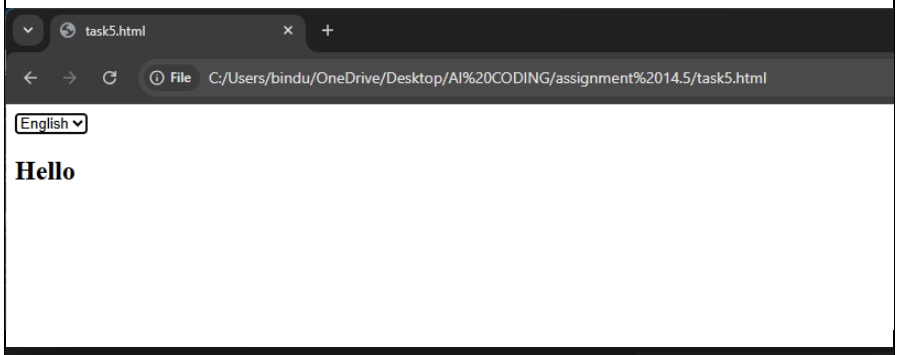
**Prompt Used**

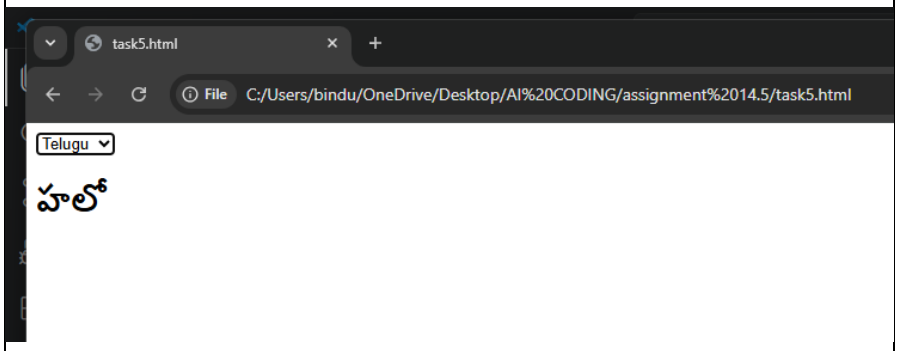
Create a multilingual website using JavaScript with language switching.

**CODE:**



**OUTPUT:**




**Explanation**

This program switches text based on selected language using JavaScript conditions.

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**Task 6 – Online Examination System**

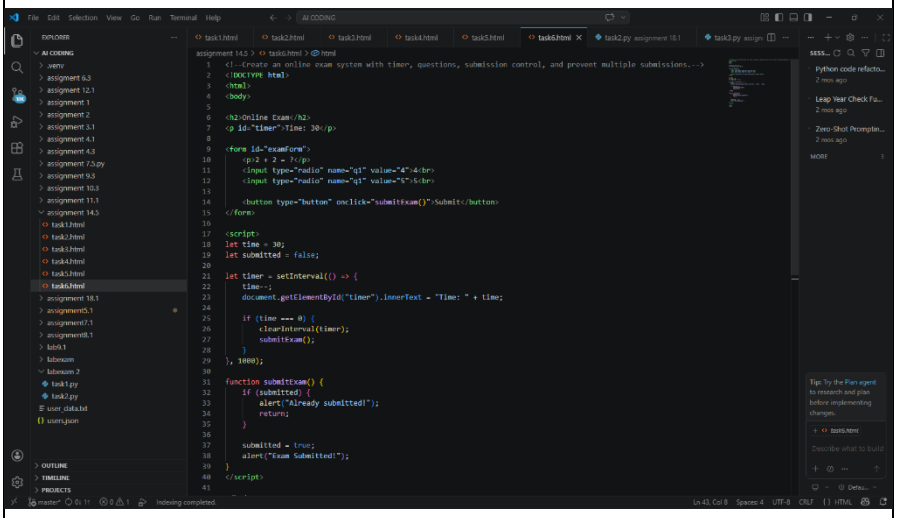
**Scenario**

Create a simple online exam with a timer.

**Prompt Used**

Create an online exam system using JavaScript with timer and submit functionality.

**Code**



```html
<!--Create an online exam system with timer, questions, submission control, and prevent multiple submissions.-->
<!DOCTYPE html>
<html>
<body>

<h2>Online Exam</h2>
<p id="timer">Time: 30</p>

<form id="examForm">
    <p>2 + 2 = ?</p>
    <input type="radio" name="q1" value="4">4<br>
    <input type="radio" name="q1" value="5">5<br>

    <button type="button" onclick="submitExam()">Submit</button>
</form>

<script>
let time = 30;
let submitted = false;

let timer = setInterval(() => {
    time--;
    document.getElementById("timer").innerText = "Time: " + time;

    if (time === 0) {
        clearInterval(timer);
        submitExam();
    }
}, 1000);

function submitExam() {
    if (submitted) {
        alert("Already submitted!");
        return;
    }

    submitted = true;
    alert("Exam Submitted!");
}
</script>
```

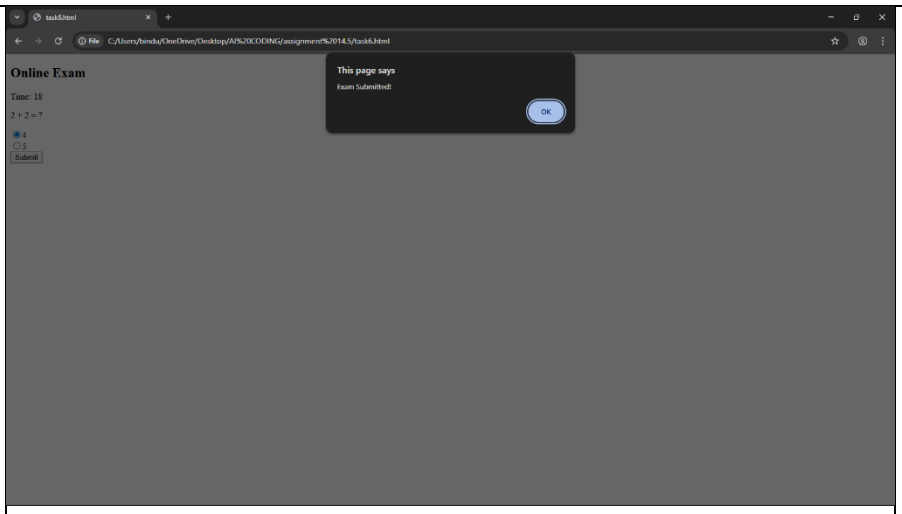
**Output**

**Online Exam**

Time: 24

2 + 2 = ?

○ 4
○ 5
Submit




**Explanation**

This program uses JavaScript timer to decrease time every second and allows submission.